# CSE 4615:: Wireless Networks

# IEEE 802.11 Wireless Local Area Networks

Lecture 5

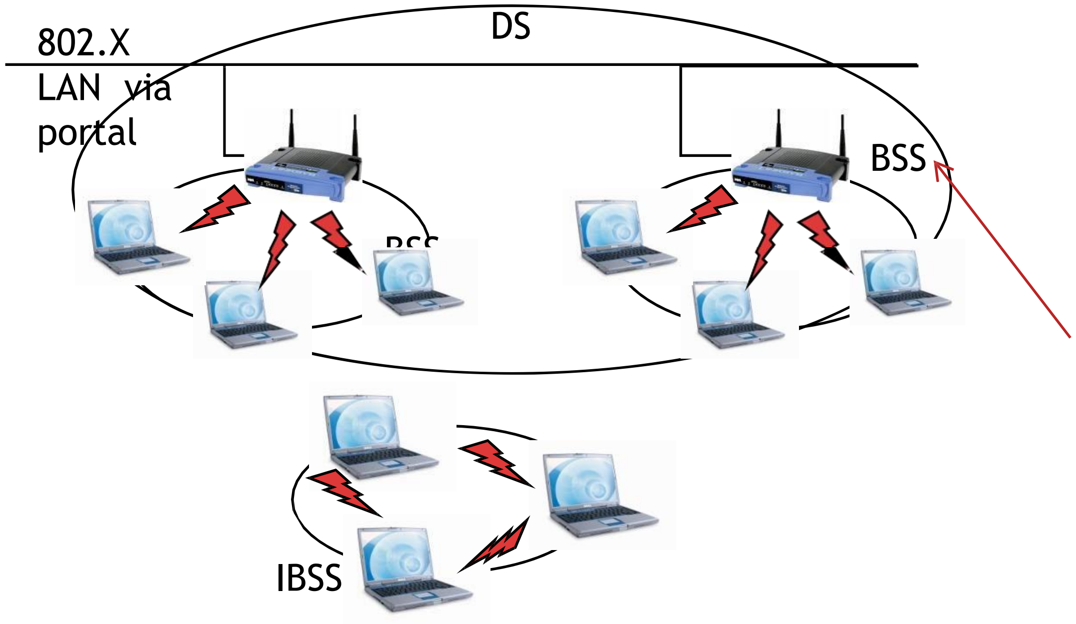
# IEEE 802.11 Wireless Local Area Networks

# Content

- Scope of 802.11
- Reference Model, Architecture, Services
- Medium Access Control Protocol

# Architecture

**802.11 Architecture & Coordination Functions**




**Basic Service Set (BSS):**
- The basic element of IEEE 802.11 Networks
- **A group of stations controlled by so-called Coordination Function (CF)**

**Coordination Function (CF):**
- **Manages the access to the wireless medium**

# Architecture

- Coordination Function (CF)
  1. The Distributed CF (DCF) is used by all stations in the BSS,
  2. whereas the Point CF (PCF) is an optional extension for the support of QoS.

  - DCF and PCF are concepts for spectrum management and medium access.

- Independent Basic Service Set (IBSS)
  - The simplest 802.11 network type. Consists of a minimum of two stations
  - No station has priority over another.
  - The responsibility of coordinating the medium access is distributed among all stations

# IEEE 802.11 Wireless LANs

802.11b
- 2.4-2.5 GHz unlicensed spectrum
- up to 11 Mbps

802.11a
- 5-6 GHz range
- up to 54 Mbps
- OFDM

802.11g
- 2.4-2.5 GHz range
- up to 54 Mbps
- OFDM

802.11n: multiple antennae
- 2.4-2.5 and 5-6 GHz range
- up to 600 Mbps
  - MIMO
  - Frame aggregation
  - Channel bonding

802.11ac
- 5-6 GHz range
- up to 3.4 Gbps
  - MU-MIMO

❖ all use CSMA/CA for multiple access

# MEDIUM ACCESS CONTROL PROTOCOL

# Listen Before Talk

- The channel sensing function is called **Clear Channel Assessment (CCA).**
  - It uses a power threshold (-82dBm, possibly implementation dependent).
  - If a station detects a signal with power larger than this threshold, the radio channel is assumed to be busy
  - Otherwise the channel is idle.
- There are different variants of CCA (see the previous Fig.)
  - Ordinary noise detection
  - Only signals from other 802.11

# Listen Before Talk

- The Network Allocation Vector (NAV) is <u>an addition to the **physical sensing**</u> of the radio channel.
- It is referred to as **virtual carrier sensing** and in fact has the function of reserving the channel for some time.
- It is a timer which decrements irrespectively of the status of the medium.
- As long as the NAV is set or the CCA sensed the radio channel as being busy, a station is not allowed to initiate transmissions.

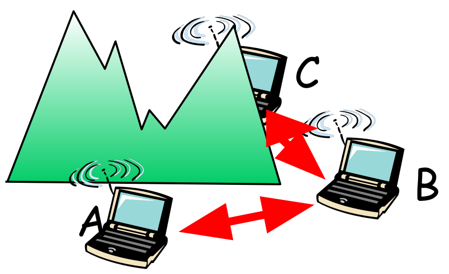
# 802.11 Medium Access Control Protocol

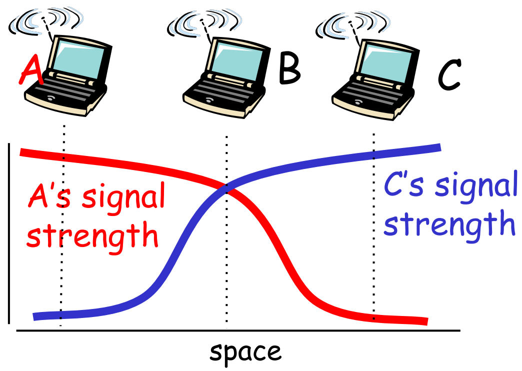
- 802.11 MAC protocol is built with the help of 2 coordination functions:
  1. DCF for traffic without QoS (asynchronous services) and
  2. PCF for traffic with QoS requirements (synchronous services).
- In the following,
  - We discuss Distributed Coordination Function (DCF)
  - an infrastructure-based BSS
- The basic 802.11 MAC protocol is the DCF that works as a **listen-before-talk** scheme, based on the Carrier Sense Multiple Access (CSMA)

# IEEE 802.11
# Distributed Coordination Function (DCF)

# Distributed Coordination Function (DCF)

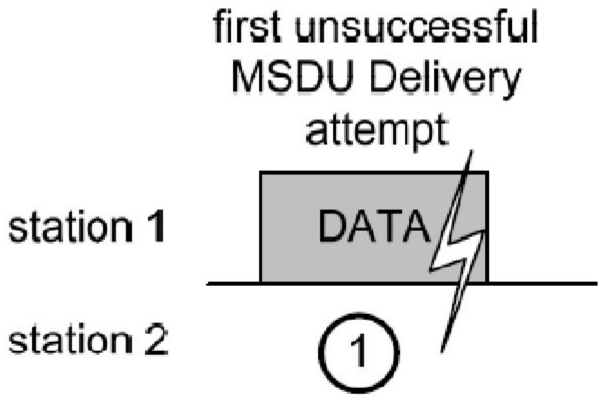
- 802.11 DCF : based on the *CSMA* - *sense before transmitting*
  - *No* collision detection (*CD*)!
    - Difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
    - Can't sense all collisions in any case: hidden terminal, fading
    - Goal: *avoid collisions:* CSMA/C(ollision)A(voidance)
  - Link layer ACKnowledgments/Retransmissions (ARQ)
    - High bit error rates



# Timing and Interframe Spaces

***Frame Exchange Scenarios***




- First attempt: DATA frame from station 1 is not received, no ACK

# Timing and Interframe Spaces

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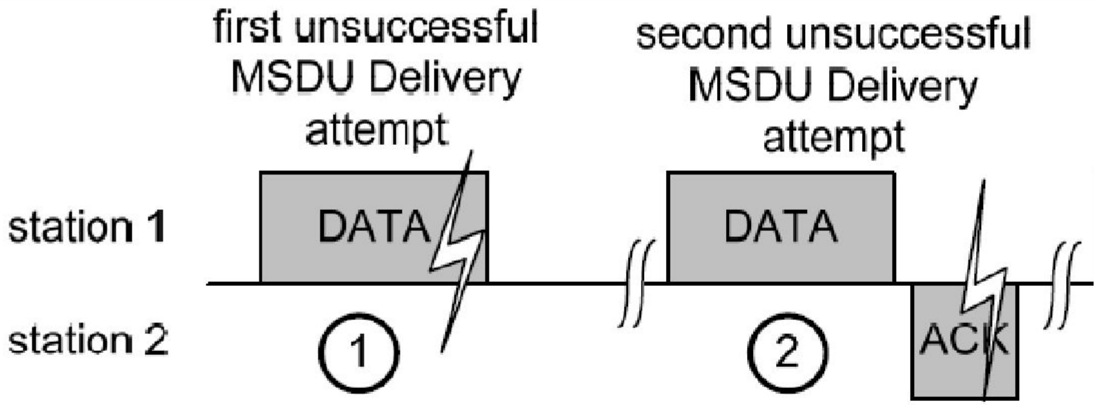
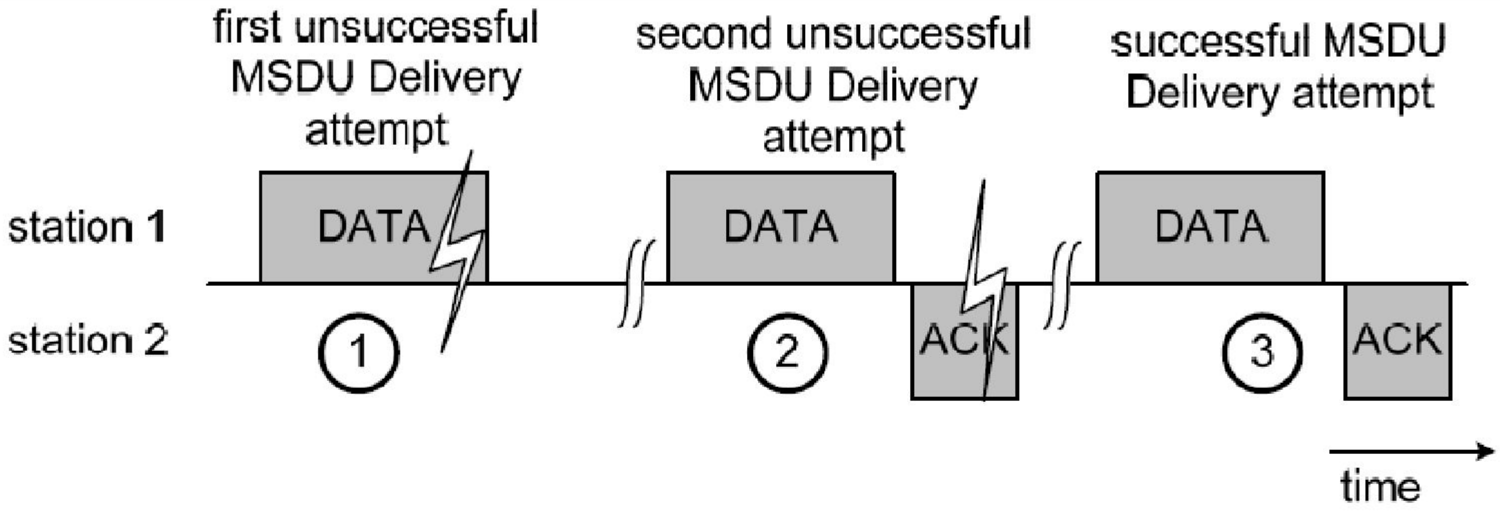



- First attempt: DATA frame from station 1 is not received, no ACK
- Second attempt: because of the missing ACK from station 2, the station 1 retransmits the DATA frame. This is successfully received, the corresponding ACK is sent as response, but missed by station 1 (for example because of interference)

# Timing and Interframe Spaces

***Frame Exchange Scenarios***




- First attempt: DATA frame from station 1 is not received, no ACK
- Second attempt: because of the missing ACK from station 2, the station 1 retransmits the DATA frame. This is successfully received, the corresponding ACK is sent as response, but missed by station 1 (for example because of interference)
- Third and successful attempt: DATA and ACK are both successfully received

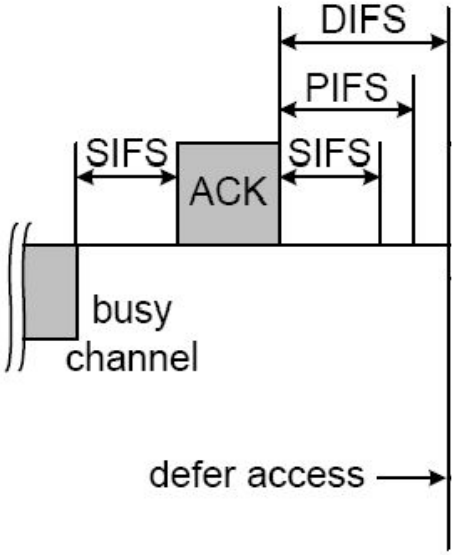
# Timing and Interframe Spaces

- The time between 2 MAC frames is called the **Interframe Space (IFS).**
- 802.11 defines 4 different IFSs:
  - Short Interframe Space (SIFS)
  - Point Coordination Function Interframe Space (PIFS)
  - Distributed Coordination Function Interframe Space (DIFS)
  - Extended Interframe Spce (EIFS)
- These IFSs represent three different levels for medium access (See the Fig. in the following)
- The shorter the IFS, the higher priority in medium access.
- All interframe spaces are independent of the channel data rate.

# Timing and Interframe Spaces

- aSlotTime
  - is used to calculate the IFSs.
  - SIFS and aSlotTime are the basis of all other duration.
  - In 802.11a, aSlotTime=9μs
- SIFS
  - is used to prioritize the immediate ACK of a data frame,
  - the response (CTS) to a RTS,
  - a subsequent MPDU of a fragmented MSDU,
  - response to any polling using the PCF, and
  - any frames of the AP during the CFP.
  - SIFS = 16μs for 802.11a
- PIFS
  - used by stations operating under the PCF to obtain channel access with highest priority.
  - PIFS = SIFS + aSlotTime.
  - PIFS = 25μs for 802.11a
- DIFS
  - used by stations operating under the DCF to obtain channel access.
  - DIFS = SIFS + 2 * aSlotTime
  - DIFS = 34μs for 802.11a

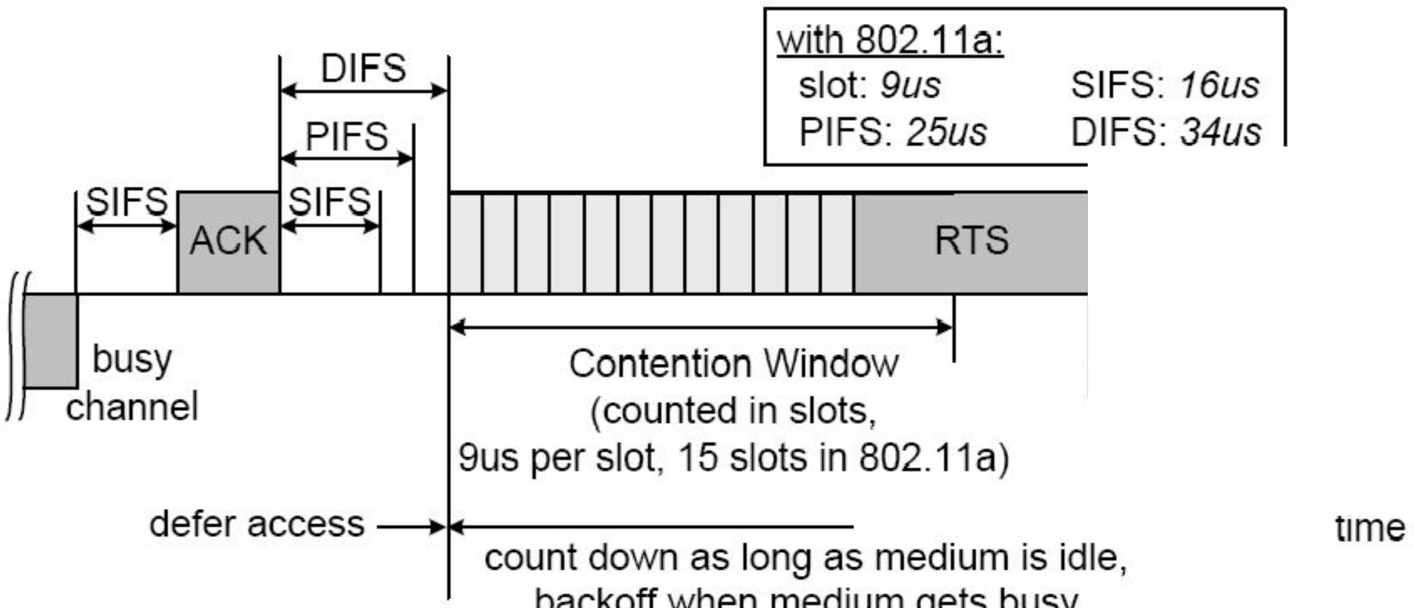
# Timing and Interframe Spaces




with 802.11a:
slot: *9us*
SIFS: *16us*
PIFS: *25us*
DIFS: *34us*

- SIFS = Short Interframe Space (no CCA carrier sensing)
- PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
- DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is too short for CCA, hence only needed for transceiver turnaround

# Timing and Interframe Spaces

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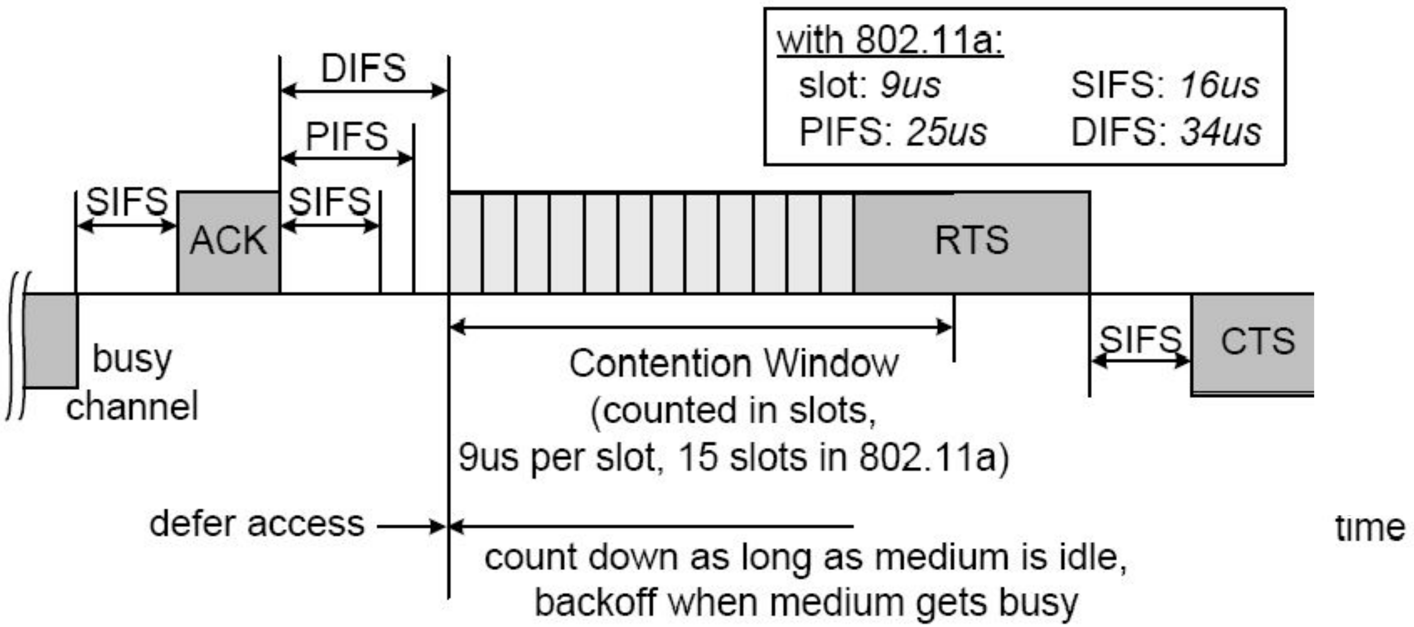


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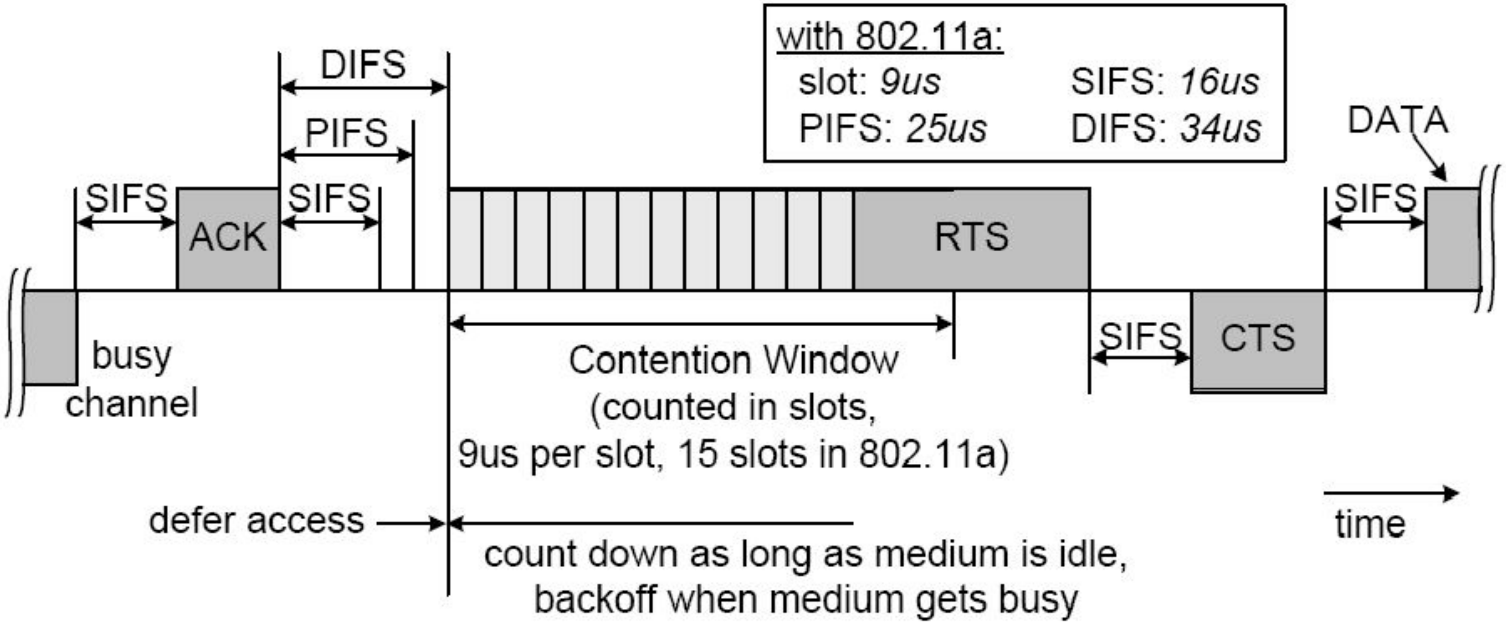


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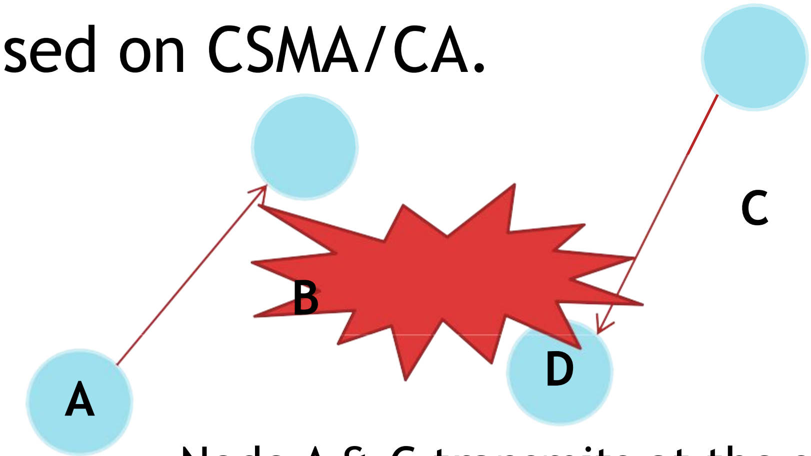
# Timing and Interframe Spaces

- EIFS (Extended IFS): used instead of DIFS when multiple stations initiated frame exchanges at different starting times, typically due to they are **hidden** from each other.
- The EIFS results in a longer deferral from channel access, which gives other stations a higher priority in medium access.
- As soon as 1 other frame is received correctly, DIFS is used again.
- EIFS is around 200μs for 802.11a

# Collision Avoidance

- A **collision** happens when there are more than 1 station attempts to transmit at the same time.
- In wireless communication, transmitter cannot detect a collision at receiver while transmitting. To account for this,
  802.11 is based on CSMA/CA.




Node A & C transmits at the same time

# Collision Avoidance

- If 2 or more stations detect the channel as being idle at the same time, inevitably a collision occurs when these stations initiate transmission at the same time.
- 802.11 defines a CA mechanism to reduce the probability of such collisions.

# Collision Avoidance

- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- A station having MSDU to deliver has to keep sensing the channel for an additional random time (=backoff) after detecting the channel as being idle for at least a DIFS (=34μs).
- Duration of Backoff =
  Random Backoff count,r x aSlotTime. (r=integer)
- Each station maintains a so-called **Contention Window (CW)**, which is used to determine n.
  - $CW = min(\ 2^{(n-1)} * CW_{min} - 1,\ \ CW_{max}\ )$
  - Random Backoff count,r = Rand(0 ~ CW)
- n= number of attempt
- In IEEE 802.11a $CW_{min} = 16$ , $CW_{max} = 1024$

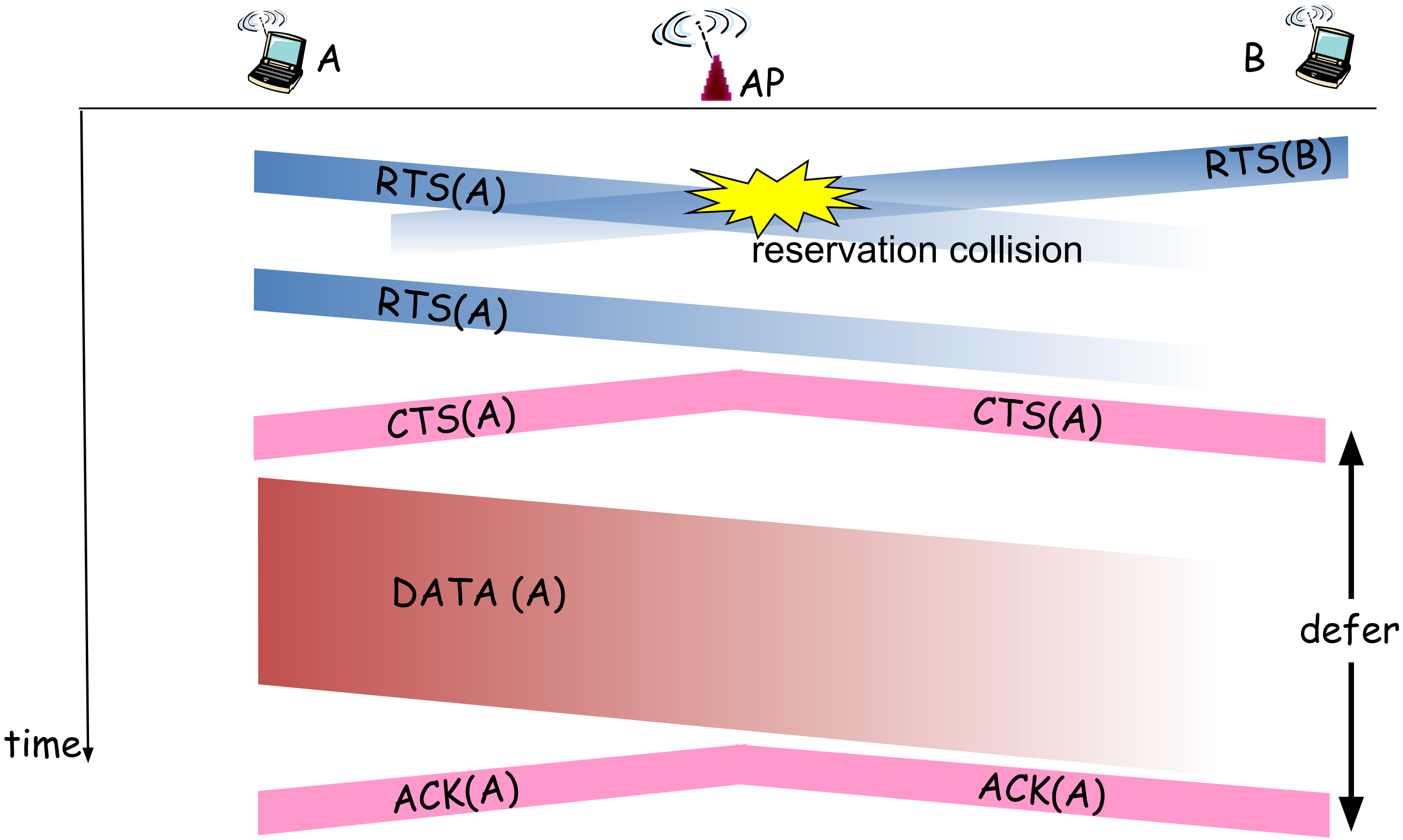
# Avoiding collisions (more)

*idea:* allow sender to "reserve" channel rather than random access of data frames: avoid collisions of long data frames

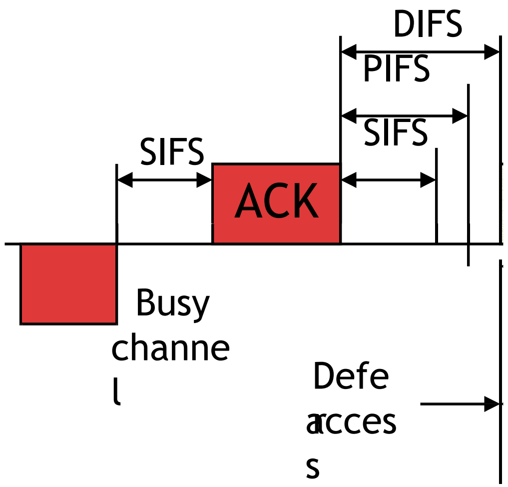
- sender first transmits *small* request-to-send (RTS) packets to BS using CSMA
  - RTSs may still collide with each other (but they're short)
- BS broadcasts clear-to-send CTS in response to RTS
- CTS heard by all nodes
  - sender transmits data frame
  - other stations defer transmissions

avoid data frame collisions completely using small reservation packets!

# Collision Avoidance: RTS-CTS exchange



# Collision Avoidance




- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- Ex: After a successful frame exchange, a station wants to send another data frame. It starts transmitting RTS after sensing the channel is idle for a duration equal to DIFS and its following backoff slots.

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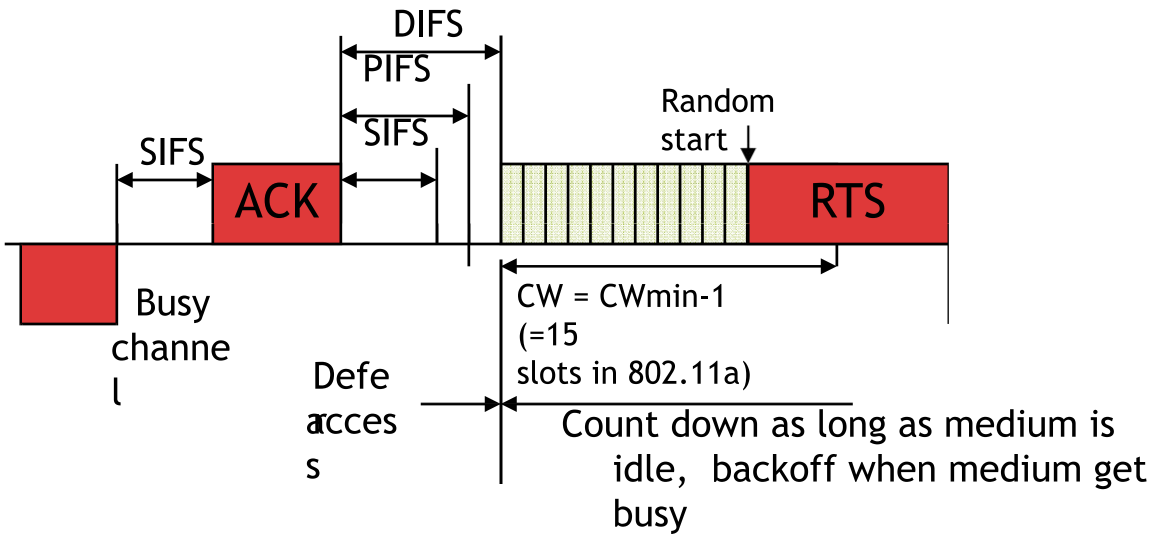


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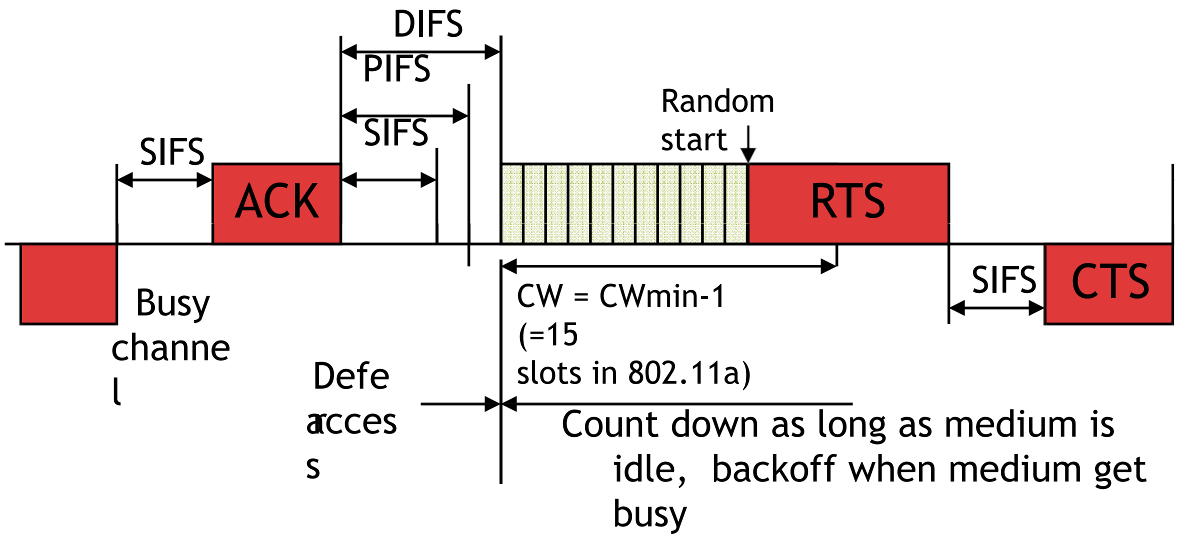


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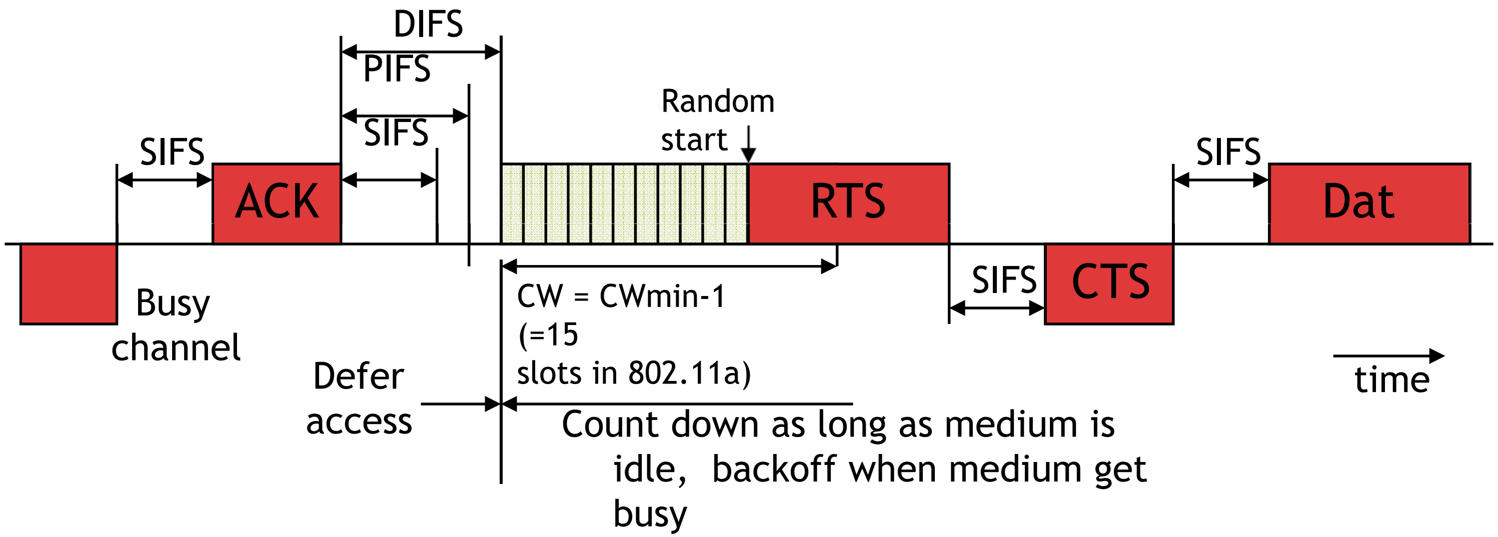


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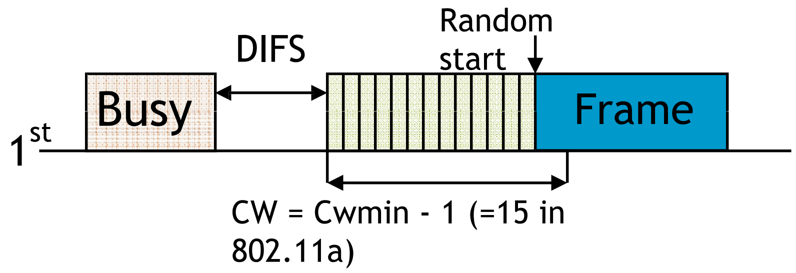


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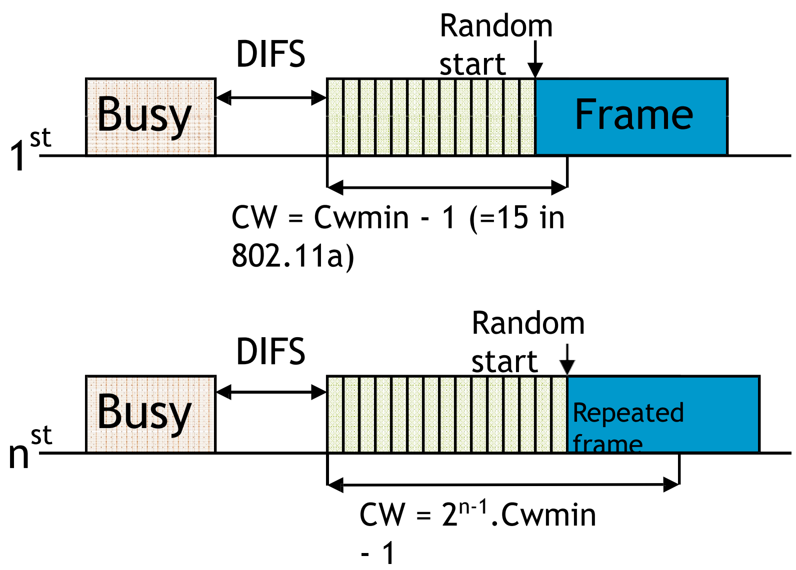
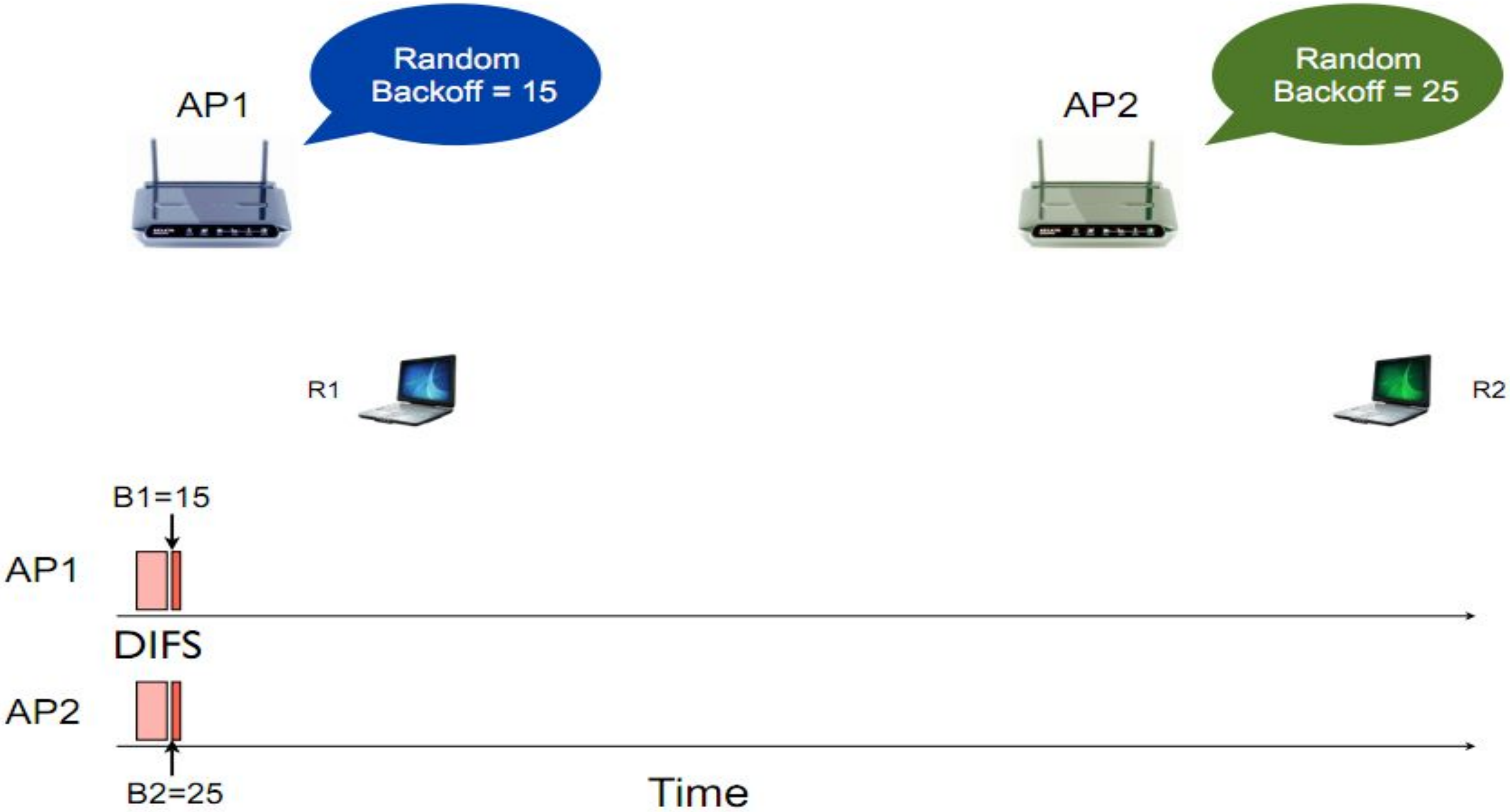



- After an unsuccessful transmission, the next backoff is performed with a doubled size of the contention window. This reduces the collision probability if there are multiple stations attempting to access the channel.

# Collision Avoidance

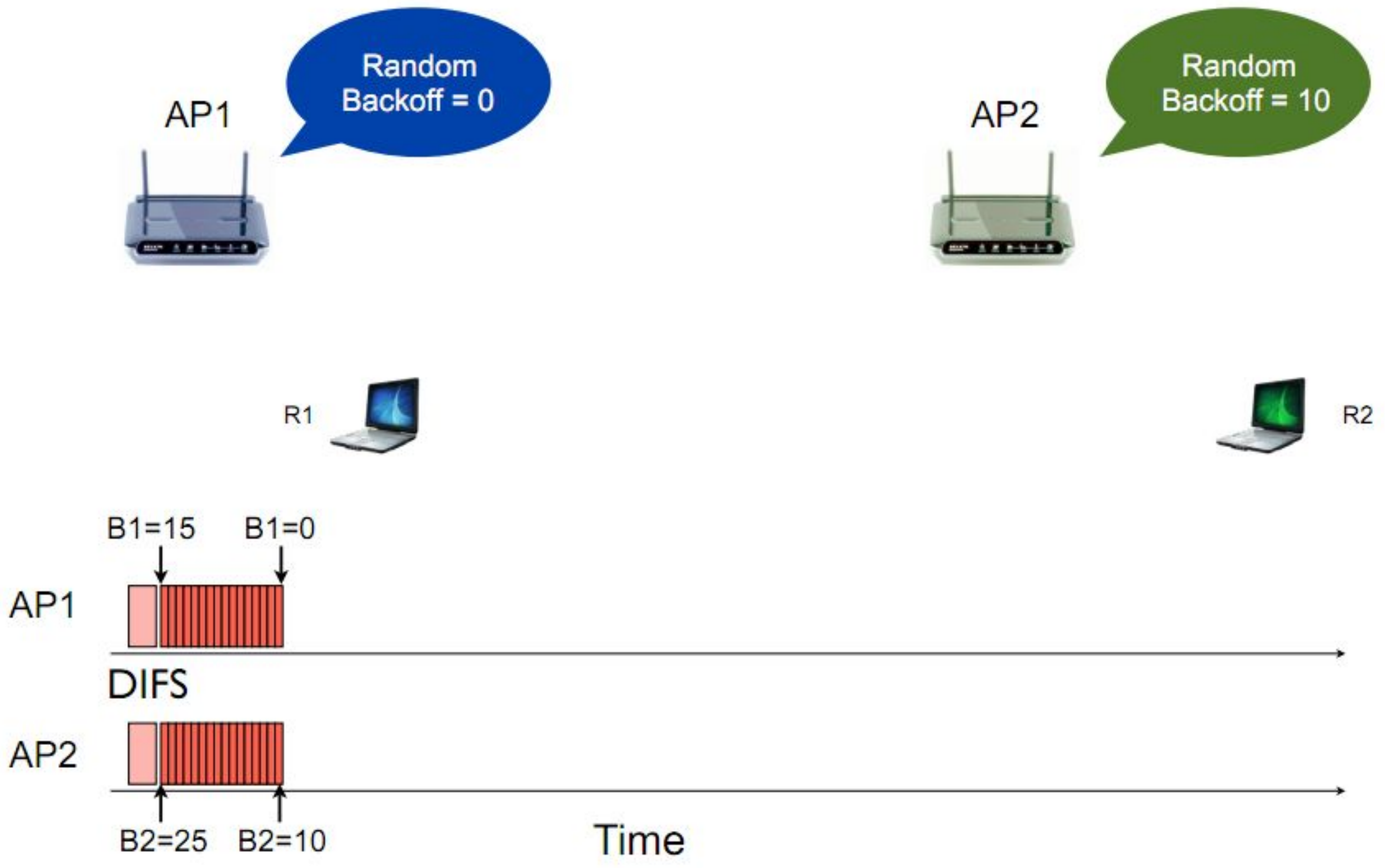
- The station that deferred from channel access during the channel busy period do not select a new random backoff time, but continue to count down the time of previous backoff after sensing the channel as idle again.
- In this way, stations, which deferred from medium access because their random backoff was larger than other's, are given a higher priority when they resume the transmission attempt.

# 802.11 Backoff Example



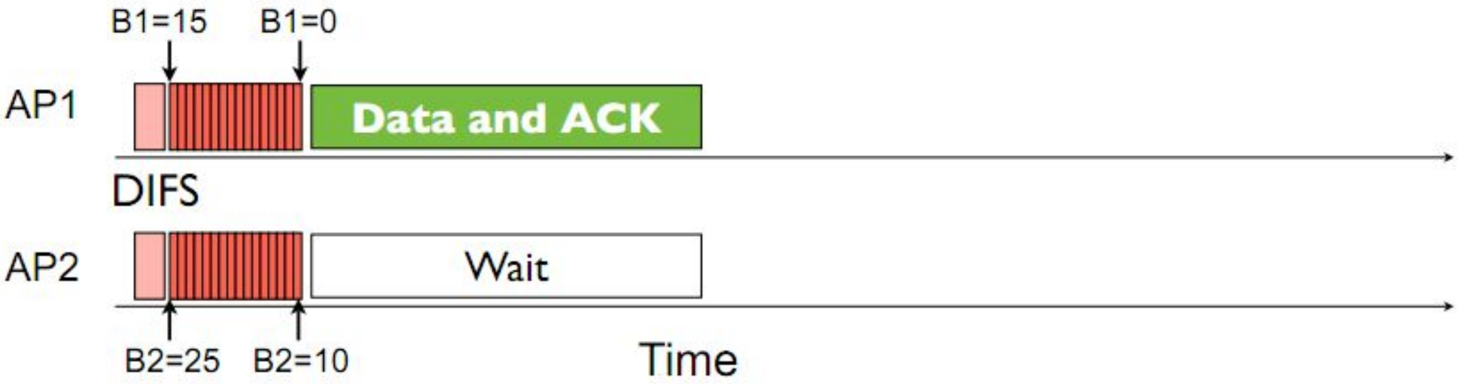
# 802.11 Backoff Example

After 15 slot countdown



# 802.11 Backoff Example


RTS, CTS Disabled Scenario !



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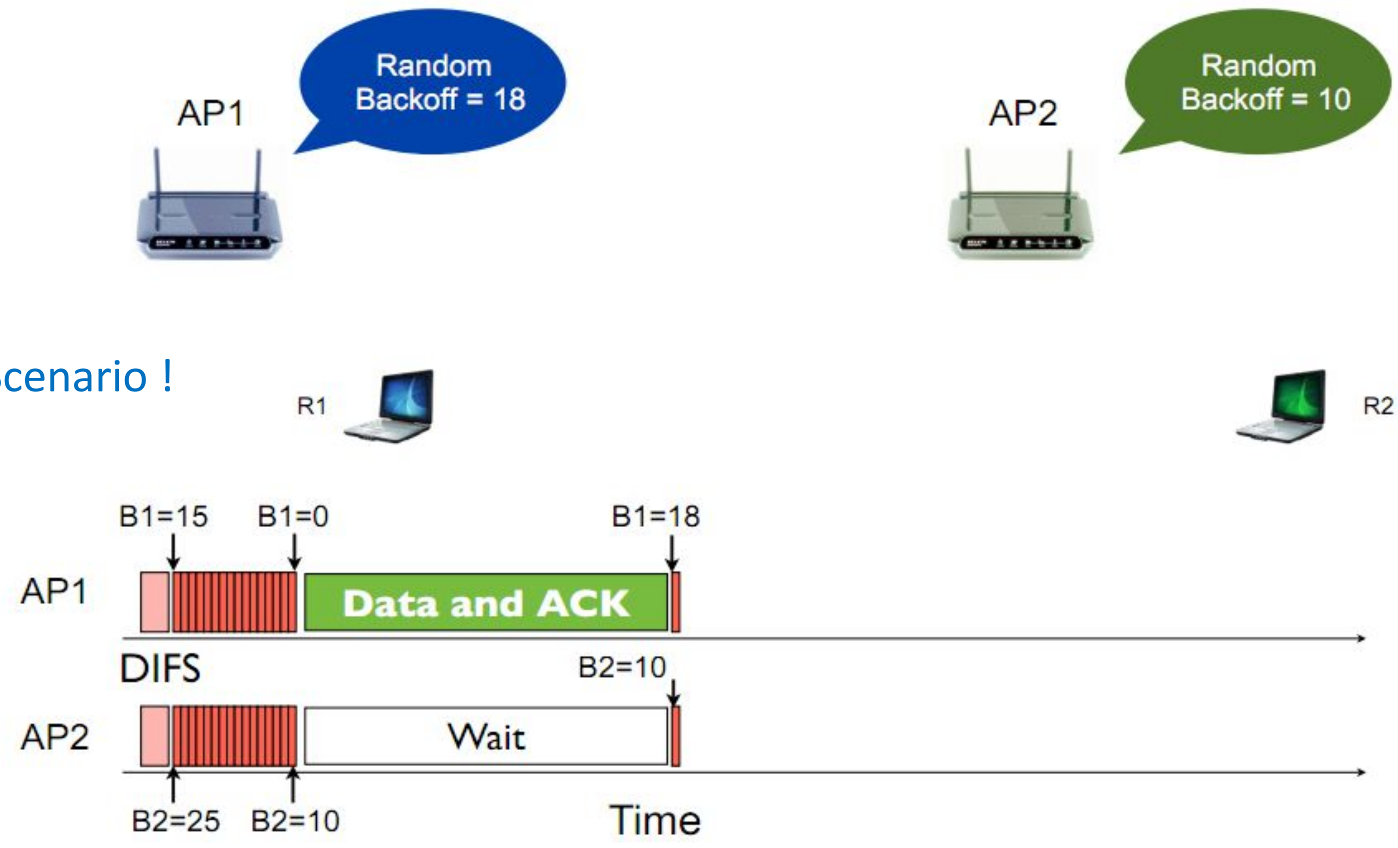

# 802.11 Backoff Example

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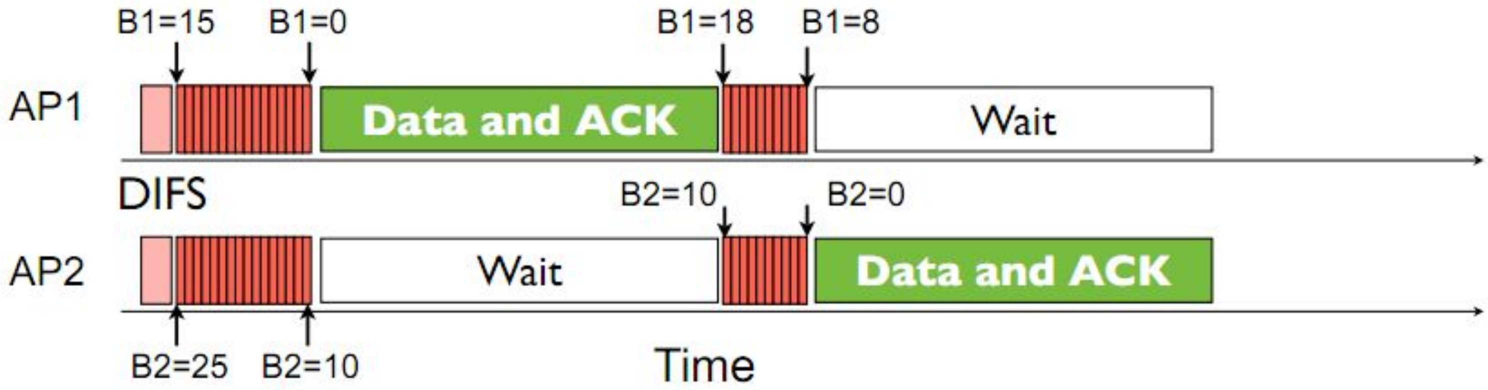


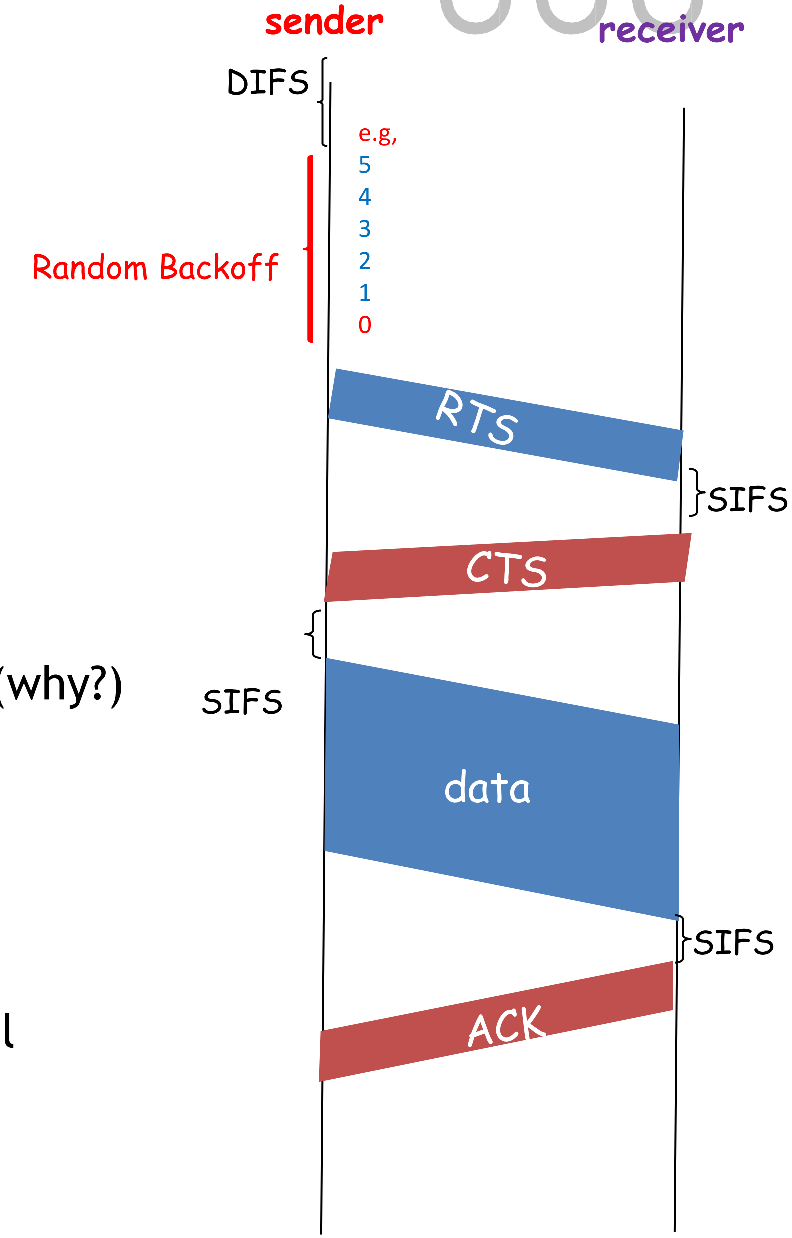
# Distributed Coordination Function (DCF)

## 802.11 sender

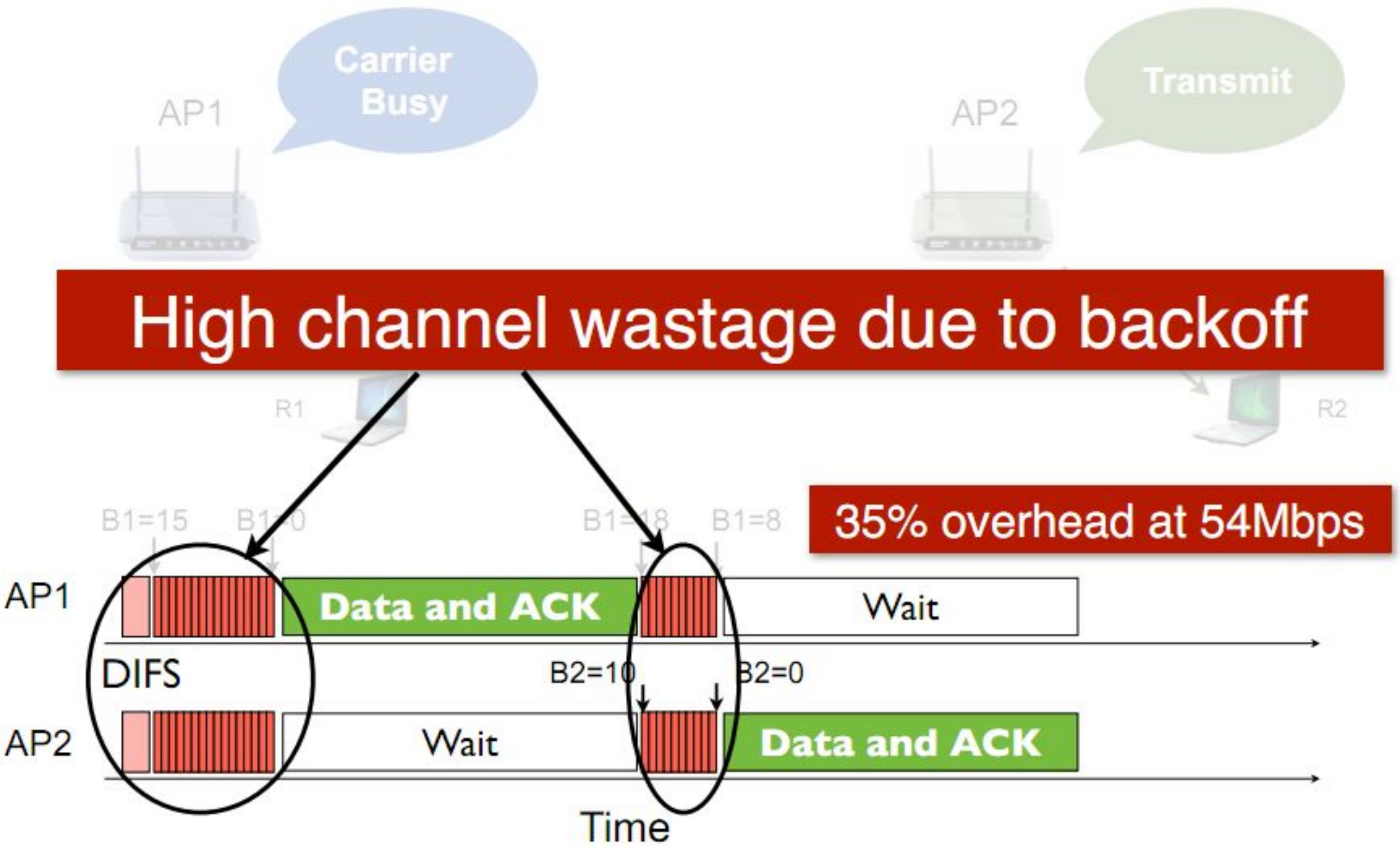
1.1 if sense channel busy then freezes
1.2 if sense channel idle for **DIFS** then
- 2.1 start random backoff time
  - timer counts down while channel idle,
  - freezes when busy,
  - resumes if idle for DIFS
  - transmit RTS when timer expires (why?)
- 3. if no CTS, then go to 1.2 (why?)
- 4. if CTS, then wait SIFS, and transmit DATA
- 5. if no ACK, increase random backoff interval, go to 1.2 (why?)

## 802.11 receiver

- if RTS frame received OK
- return CTS after **SIFS**
- if Data frame received OK
- return ACK after **SIFS** (ACK needed due to hidden terminal problem)



# 802.11 Medium Access Overhead



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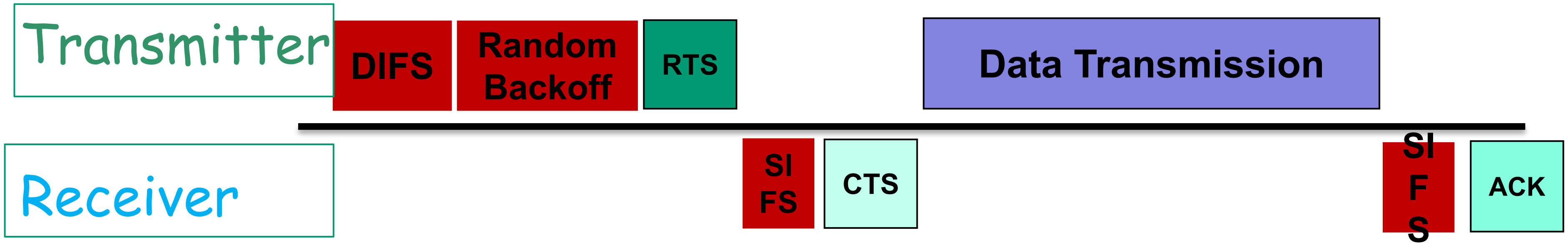



- Overhead:
  - DIFS
  - Random backoff
  - ACK/SIFS
  - Optional RTS/CTS handshake before transmission of data packet (often disabled due to its overhead)
  - Header overhead
- 802.11 has room for improvement. How?
  - (Will be discussed in coming lectures)

# 802.11 Medium Access Overhead


High channel wastage due to **Collision**

# 802.11 Medium Access Overhead

**Significant Inefficiencies of 802.11 DCF :**

- **Idle Channel Time:** During the countdown, the channel remains idle, even though no transmission occurs. This results in underutilization of available bandwidth.
- **Collision Probability:** When two or more nodes pick the same backoff value, simultaneous transmission leads to **packet collisions**. These collisions cause retransmissions and an **exponential increase in CW**, further delaying future transmissions.

- **Exponential Backoff Under Congestion:** In IEEE 802.11, Binary Exponential Backoff (BEB) increases CW size as:
  - *CW = min(2^n * CWmin - 1,* Cwmax)
- for each failed attempt (up to CWmax = 1024).
- At 10 active nodes, probability of at least two nodes choosing the same slot exceeds 40%, drastically reducing throughput and increasing latency.

# 802.11 Medium Access Overhead


High channel wastage due to exponential backoff

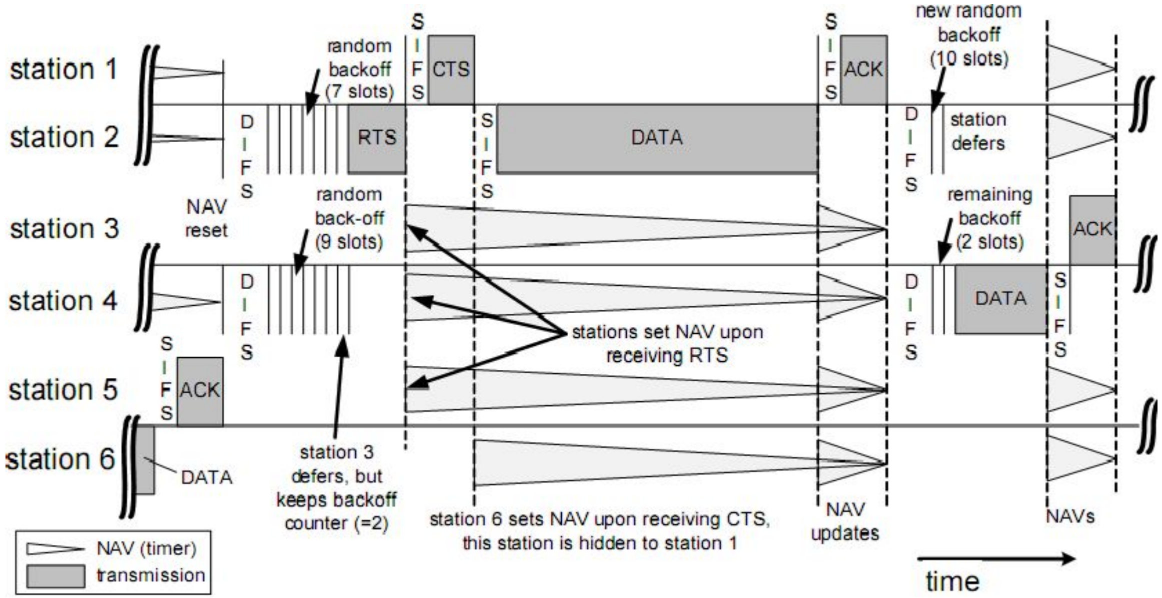
# Hidden Stations and RTS/CTS

- Hidden Stations problem arises when a station is able to successfully receive frames from two different stations but the two stations cannot detect each other.
- To reduce throughput reduction owing to this problem, 802.11 specifies exchange of Request-to- Send/Clear-to-Send (**RTS/CTS**) frames as an option made by transmitting station.
- Consecutive frames in the sequence of RTS, CTS, data and ACK are spaced by an SIFS duration for transceiver turn-around.

# Hidden Stations and RTS/CTS

- The RTS/CTS carry in its duration fields the information, how long the sequence RTS, CTS, Data, ACK will take.
- Hence, other stations close to the transmitting station and hidden stations close to the receiving station will not start any transmissions; their NAV timer is set.
- A hidden station close to receiving station might not receive the RTS due to the large distance, but will in most cases receive the CTS frame

# Hidden Stations and RTS/CTS




- Station 6 cannot detect the RTS from station 2, but can hear the CTS of station 1. Although station 6 is hidden to station 2, it refrains from channel access because of NAV.
- Note that SIFS is shorter than DIFS, which always gives CTS and ACK the highest priority for access to the channel